

Quadratic Inequalities

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Quadratic Inequalities (ref ise4)

This document contains all 50 *Quadratic Inequalities* questions, each with a fully worked solution and a TikZ sketch of the parabola. The roots are marked on the x -axis and the solution set is highlighted in orange (open circles for strict $</>$, filled for $\leq/\geq$). Threshold lines are drawn for the double-inequality and modelling questions, and a family of parabolas illustrates the discriminant (k/p) questions.

Contents

Quadratic Inequalities 01 (ise4-001)	1
Quadratic Inequalities 02 (ise4-002)	2
Quadratic Inequalities 03 (ise4-003)	3
Quadratic Inequalities 04 (ise4-004)	4
Quadratic Inequalities 05 (ise4-005)	5
Quadratic Inequalities 06 (ise4-006)	6
Quadratic Inequalities 07 (ise4-007)	7
Quadratic Inequalities 08 (ise4-008)	8
Quadratic Inequalities 09 (ise4-009)	8
Quadratic Inequalities 10 (ise4-010)	9
Quadratic Inequalities 11 (ise4-011)	10
Quadratic Inequalities 12 (ise4-012)	11
Quadratic Inequalities 13 (ise4-013)	12
Quadratic Inequalities 14 (ise4-014)	13
Quadratic Inequalities 15 (ise4-015)	14

Quadratic Inequalities 16 (ise4-016)	14
Quadratic Inequalities 17 (ise4-017)	15
Quadratic Inequalities 18 (ise4-018)	16
Quadratic Inequalities 19 (ise4-019)	17
Quadratic Inequalities 20 (ise4-020)	18
Quadratic Inequalities 21 (ise4-021)	19
Quadratic Inequalities 22 (ise4-022)	20
Quadratic Inequalities 23 (ise4-023)	21
Quadratic Inequalities 24 (ise4-024)	22
Quadratic Inequalities 25 (ise4-025)	23
Quadratic Inequalities 26 (ise4-026)	24
Quadratic Inequalities 27 (ise4-027)	25
Quadratic Inequalities 28 (ise4-028)	26
Quadratic Inequalities 29 (ise4-029)	27
Quadratic Inequalities 30 (ise4-030)	28
Quadratic Inequalities 31 (ise4-031)	29
Quadratic Inequalities 32 (ise4-032)	30
Quadratic Inequalities 33 (ise4-033)	31
Quadratic Inequalities 34 (ise4-034)	32
Quadratic Inequalities 35 (ise4-035)	33
Quadratic Inequalities 36 (ise4-036)	34
Quadratic Inequalities 37 (ise4-037)	35
Quadratic Inequalities 38 (ise4-038)	36
Quadratic Inequalities 39 (ise4-039)	37

Quadratic Inequalities 40 (ise4-040)	38
Quadratic Inequalities 41 (ise4-041)	39
Quadratic Inequalities 42 (ise4-042)	40
Quadratic Inequalities 43 (ise4-043)	41
Quadratic Inequalities 44 (ise4-044)	42
Quadratic Inequalities 45 (ise4-045)	43
Quadratic Inequalities 46 (ise4-046)	44
Quadratic Inequalities 47 (ise4-047)	45
Quadratic Inequalities 48 (ise4-048)	46
Quadratic Inequalities 49 (ise4-049)	47
Quadratic Inequalities 50 (ise4-050)	48

Quadratic Inequalities 01 (ise4-001)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $x^2 - 5x + 6 > 0$. Give your answer in set notation.

Worked solution.

Step 1: Factorise the quadratic.

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

We look for two numbers that multiply to 6 and add to -5; these are -2 and -3. The roots are $x = 2$ and $x = 3$.

Step 2: Analyse the sign of the parabola.

$$a = 1 > 0 \implies \text{parabola opens upward}$$

Because the leading coefficient is positive, the parabola is U-shaped. It is negative BETWEEN the roots and positive OUTSIDE the roots. We want it to be strictly positive.

Step 3: Read off the values of x for which the parabola is above the x -axis.

$$x < 2 \quad \text{or} \quad x > 3$$

The strict inequality > 0 excludes the roots themselves (where the curve equals 0).

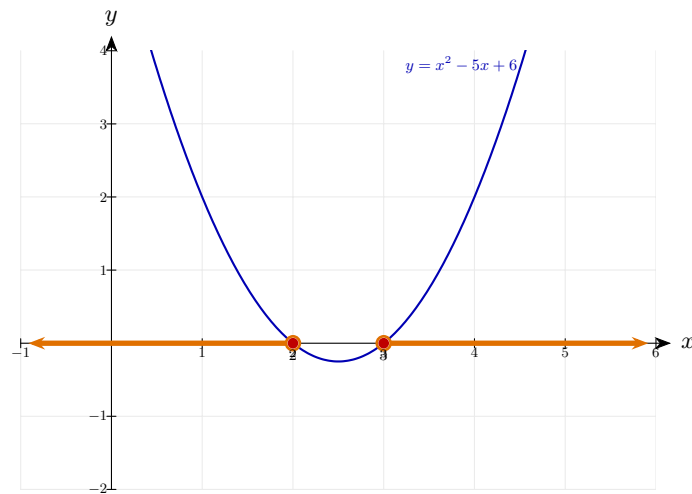
Step 4: State the answer in set notation.

$$\{x : x < 2\} \cup \{x : x > 3\}$$

A union of two disjoint open intervals.

Final answer: $\{x : x < 2\} \cup \{x : x > 3\}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 02 (ise4-002)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $x^2 - 7x + 12 < 0$. Give your answer in interval notation.

Worked solution.

Step 1: Factorise the quadratic.

$$x^2 - 7x + 12 = (x - 3)(x - 4)$$

Two numbers multiplying to 12 and summing to -7 are -3 and -4. Roots: $x = 3$, $x = 4$.

Step 2: Analyse the sign of the parabola.

$$a = 1 > 0 \implies \text{U-shape; negative between the roots}$$

For $a > 0$ the quadratic is below the x -axis between the two roots. We want where the expression is strictly less than zero, so we take the region between the roots, excluding the endpoints.

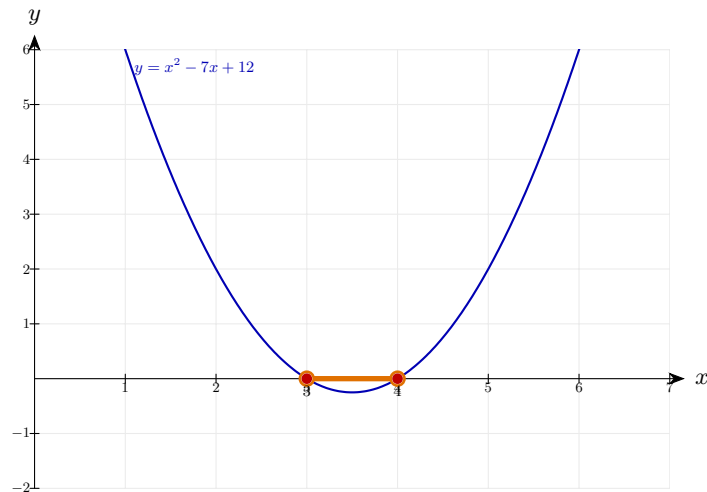
Step 3: Write the solution as an open interval.

$$3 < x < 4$$

In interval notation this is $(3, 4)$, with round brackets because the inequality is strict.

Final answer: $x \in (3, 4)$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 03 (ise4-003)

Difficulty: Foundation / Marks: 3

Solve $x^2 + 2x - 8 > 0$. Give your answer in set notation.

Worked solution.

Step 1: Factorise.

$$x^2 + 2x - 8 = (x + 4)(x - 2)$$

Two numbers multiplying to -8 and summing to 2 are 4 and -2. Roots: $x = -4$, $x = 2$.

Step 2: Identify the shape of the parabola.

$$a = 1 > 0 \implies \text{U-shape; positive outside the roots}$$

A positive leading coefficient means a U-shaped parabola: it is below the x -axis between the roots and above the x -axis to the left of the smaller root and to the right of the larger root.

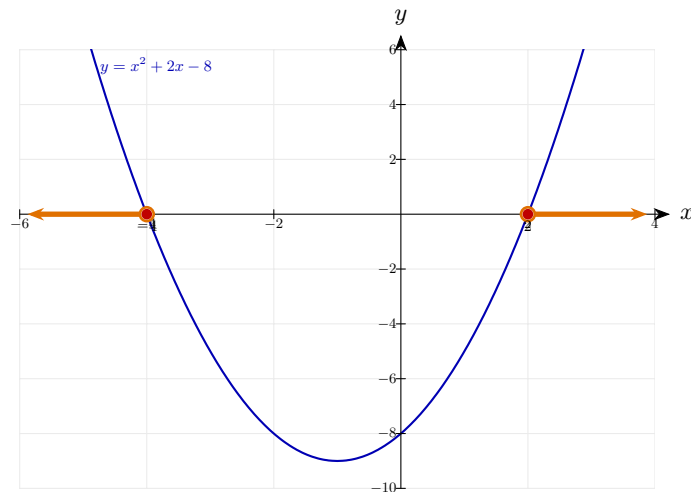
Step 3: Read off the solution.

$$x < -4 \quad \text{or} \quad x > 2$$

Strict inequality, so the roots themselves are excluded.

Final answer: $\{x : x < -4\} \cup \{x : x > 2\}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 04 (ise4-004)

Difficulty: Foundation / Marks: 3

Solve $x^2 - x - 12 < 0$. Give your answer in interval notation.

Worked solution.

Step 1: Factorise.

$$x^2 - x - 12 = (x - 4)(x + 3)$$

Two numbers multiplying to -12 and summing to -1 are -4 and 3. Roots: $x = -3$, $x = 4$.

Step 2: Sign analysis.

$$a = 1 > 0 \implies \text{quadratic is negative between the roots}$$

U-shaped parabola, so the expression is below the x -axis on the interval between the two roots.

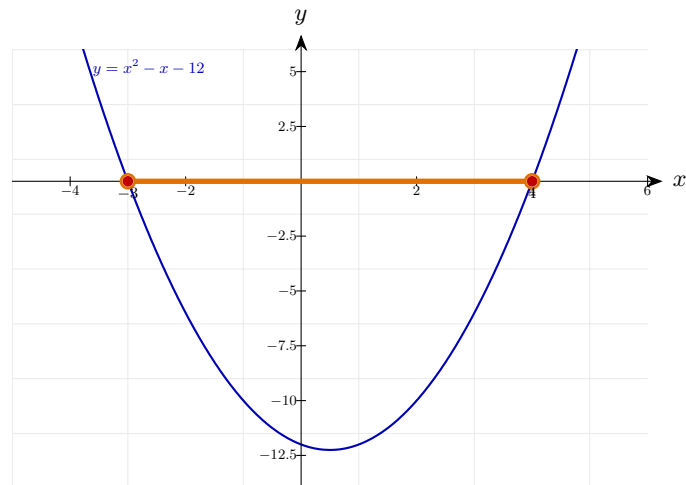
Step 3: State the solution interval.

$$-3 < x < 4$$

Open interval because the inequality is strict.

Final answer: $x \in (-3, 4)$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 05 (ise4-005)

Difficulty: Foundation / Marks: 3

Find the set of values of x for which $x^2 - 9 > 0$.

Worked solution.

Step 1: Factorise using the difference of two squares.

$$x^2 - 9 = (x - 3)(x + 3)$$

$a^2 - b^2 = (a - b)(a + b)$. Roots are $x = \pm 3$.

Step 2: Identify the shape and sign regions.

$$a = 1 > 0 \implies \text{U-shape; positive outside the roots}$$

A U-shape symmetric about the y -axis. The quadratic is positive where x is further from 0 than 3, and negative for $-3 < x < 3$.

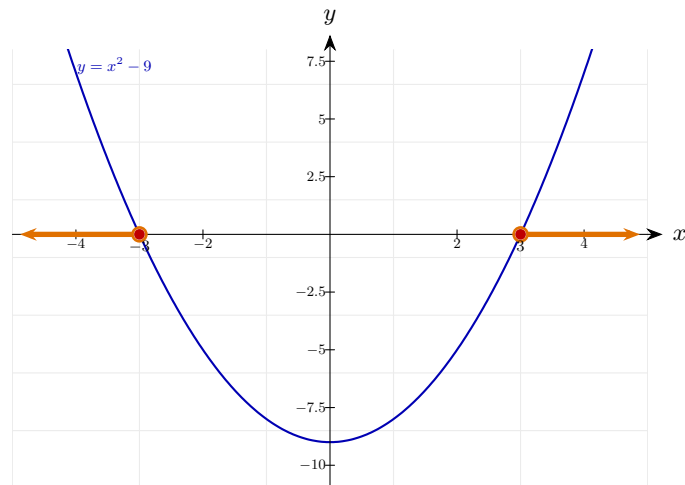
Step 3: State the solution.

$$x < -3 \quad \text{or} \quad x > 3$$

Strict inequality, so $x = \pm 3$ are excluded.

Final answer: $x < -3$ or $x > 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 06 (ise4-006)

Difficulty: Foundation / Marks: 3

Solve $x^2 + 6x + 5 < 0$.

Worked solution.

Step 1: Factorise.

$$x^2 + 6x + 5 = (x + 1)(x + 5)$$

Two numbers multiplying to 5 and summing to 6 are 1 and 5. Roots: $x = -1$, $x = -5$.

Step 2: Determine sign behaviour.

$$a = 1 > 0 \implies \text{U-shape; negative between the roots}$$

The expression is below zero for x between -5 and -1 .

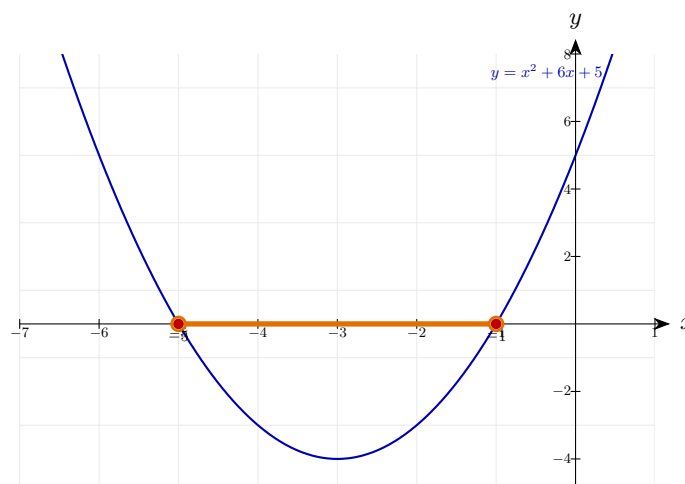
Step 3: Write the solution.

$$-5 < x < -1$$

Both endpoints excluded because of the strict inequality.

Final answer: $-5 < x < -1$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 07 (ise4-007)

Difficulty: Foundation / Marks: 4

Solve $-x^2 + 4x + 5 > 0$.

Worked solution.

Step 1: Multiply through by -1 , reversing the inequality.

$$-x^2 + 4x + 5 > 0 \iff x^2 - 4x - 5 < 0$$

Multiplying an inequality by a negative number flips the direction. This is the most reliable way to handle a negative leading coefficient — convert it to a monic problem you already know how to solve.

Step 2: Factorise the monic form.

$$x^2 - 4x - 5 = (x - 5)(x + 1)$$

Two numbers multiplying to -5 and summing to -4 are -5 and 1 . Roots: $x = -1$, $x = 5$.

Step 3: Read off the solution from the monic inequality.

$$(x - 5)(x + 1) < 0 \implies -1 < x < 5$$

The monic parabola is U-shaped and lies below the x -axis between its roots.

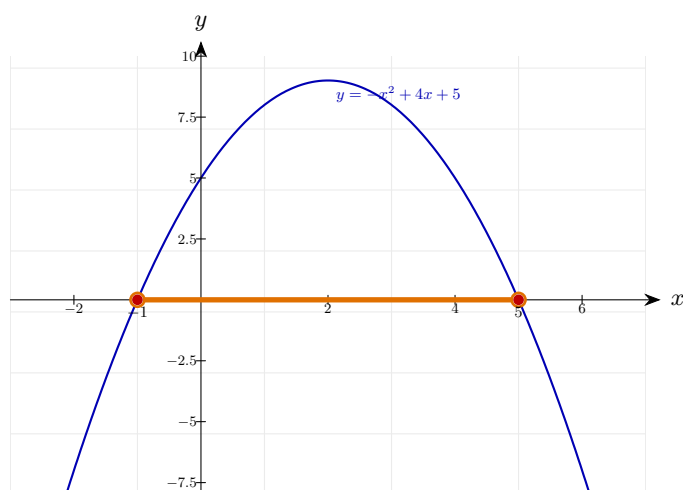
Step 4: Sanity check using the original parabola.

$$a = -1 < 0 \implies \text{n-shape; positive BETWEEN the roots}$$

For $a < 0$ the parabola opens downward, so the expression is positive between the roots — opposite to the monic case. Both methods agree.

Final answer: $-1 < x < 5$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 08 (ise4-008)

Difficulty: Foundation / Marks: 4

Solve $-x^2 + 2x + 3 < 0$. Give your answer in set notation.

Worked solution.

Step 1: Multiply through by -1 and flip the inequality.

$$-x^2 + 2x + 3 < 0 \iff x^2 - 2x - 3 > 0$$

Reverse the inequality sign whenever you multiply or divide by a negative quantity. This is the most common error students make on these problems.

Step 2: Factorise.

$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

Roots: $x = -1$, $x = 3$.

Step 3: Apply the U-shape rule to the monic form.

$$(x - 3)(x + 1) > 0 \implies x < -1 \text{ or } x > 3$$

A positive-leading-coefficient parabola is positive outside its roots.

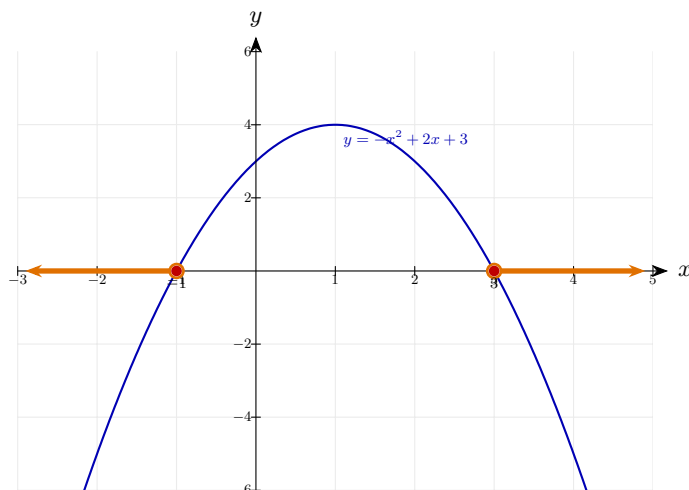
Step 4: Write in set notation.

$$\{x : x < -1\} \cup \{x : x > 3\}$$

Union of the two intervals where the original expression is below zero.

Final answer: $\{x : x < -1\} \cup \{x : x > 3\}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 09 (ise4-009)

Difficulty: Foundation / Marks: 4

Solve $6 + x - x^2 > 0$.

Worked solution.

Step 1: Rewrite in standard form and multiply through by -1 .

$$-x^2 + x + 6 > 0 \iff x^2 - x - 6 < 0$$

First arrange terms in descending powers of x ; then flip the inequality on multiplying by -1 .

Step 2: Factorise.

$$x^2 - x - 6 = (x - 3)(x + 2)$$

Roots: $x = -2$, $x = 3$.

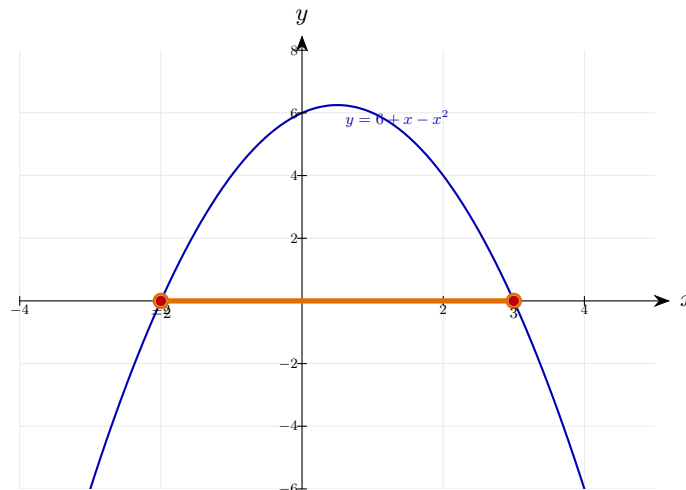
Step 3: Use the U-shape rule.

$$(x - 3)(x + 2) < 0 \implies -2 < x < 3$$

Below zero between the roots for a U-shaped ($a > 0$) parabola.

Final answer: $-2 < x < 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 10 (ise4-010)

Difficulty: Foundation / Marks: 4

Solve $12 - 4x - x^2 < 0$.

Worked solution.

Step 1: Rewrite in standard form and multiply through by -1 .

$$-x^2 - 4x + 12 < 0 \iff x^2 + 4x - 12 > 0$$

Flip the inequality when multiplying by -1 . Now we have a monic problem.

Step 2: Factorise.

$$x^2 + 4x - 12 = (x + 6)(x - 2)$$

Roots: $x = -6$, $x = 2$.

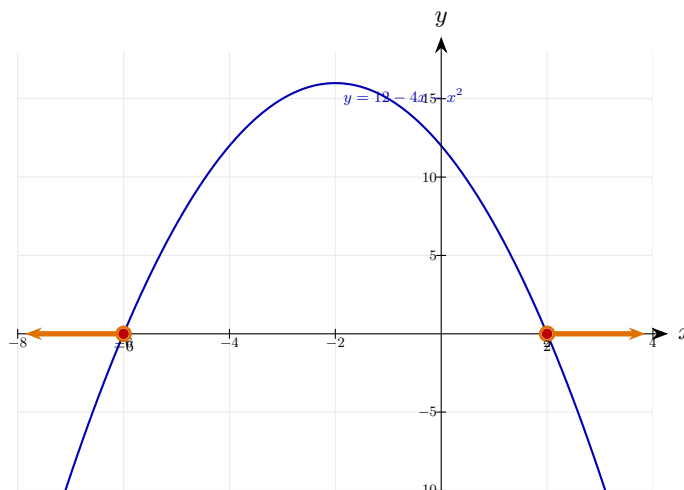
Step 3: Apply the U-shape rule.

$$(x + 6)(x - 2) > 0 \implies x < -6 \text{ or } x > 2$$

Positive outside the roots for a U-shaped parabola.

Final answer: $x < -6$ or $x > 2$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 11 (ise4-011)

Difficulty: Foundation / Marks: 3

Solve $x^2 - 6x + 8 \leq 0$.

Worked solution.

Step 1: Factorise.

$$x^2 - 6x + 8 = (x - 2)(x - 4)$$

Roots: $x = 2$, $x = 4$.

Step 2: Determine where the quadratic is non-positive.

$$a = 1 > 0 \implies \text{negative between, zero at, the roots}$$

A U-shaped parabola sits below or on the x -axis on the closed interval between its roots.

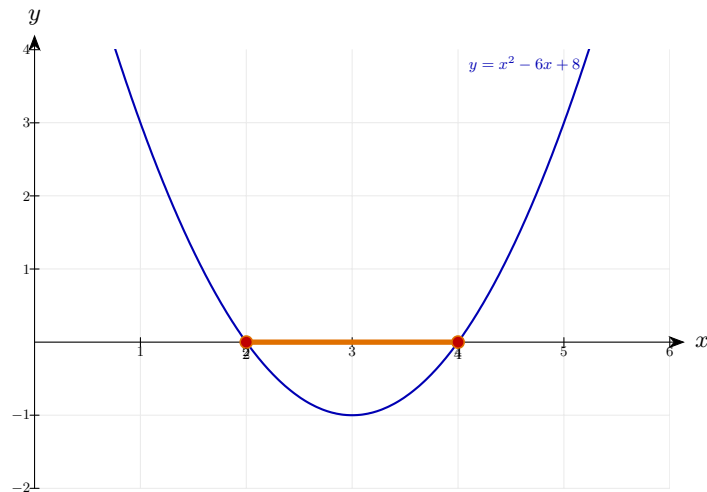
Step 3: Include the endpoints because of $\leq$.

$$2 \leq x \leq 4$$

At the roots the quadratic equals 0, which satisfies ≤ 0 . So the endpoints belong in the solution; use a closed interval $[2, 4]$.

Final answer: $2 \leq x \leq 4$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 12 (ise4-012)

Difficulty: Foundation / Marks: 3

Solve $x^2 + x - 6 \geq 0$. Give your answer in interval notation.

Worked solution.

Step 1: Factorise.

$$x^2 + x - 6 = (x + 3)(x - 2)$$

Roots: $x = -3$, $x = 2$.

Step 2: Apply the U-shape rule with inclusive endpoints.

$$a = 1 > 0 \implies \text{non-negative outside / on the roots}$$

A U-shaped parabola is at or above the x -axis when $x \leq$ smaller root or $x \geq$ larger root. We include the endpoints because the inequality is non-strict.

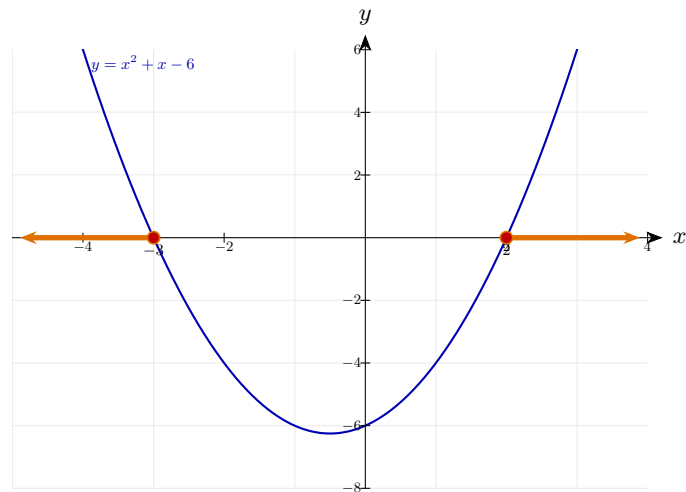
Step 3: Write the solution as a union of closed intervals.

$$x \leq -3 \text{ or } x \geq 2$$

In interval notation: $(-\infty, -3] \cup [2, \infty)$. Square brackets denote inclusive endpoints; $\pm\infty$ always takes a round bracket.

Final answer: $x \in (-\infty, -3] \cup [2, \infty)$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 13 (ise4-013)

Difficulty: Foundation / Marks: 3

Solve $x^2 - 4 \leq 0$.

Worked solution.

Step 1: Factorise using the difference of two squares.

$$x^2 - 4 = (x - 2)(x + 2)$$

Roots: $x = \pm 2$.

Step 2: Determine the sign of the parabola.

$$a = 1 > 0 \implies \text{U-shape; } \leq 0 \text{ between roots (inclusive)}$$

Includes endpoints because the quadratic equals 0 at $x = \pm 2$, which satisfies ≤ 0 .

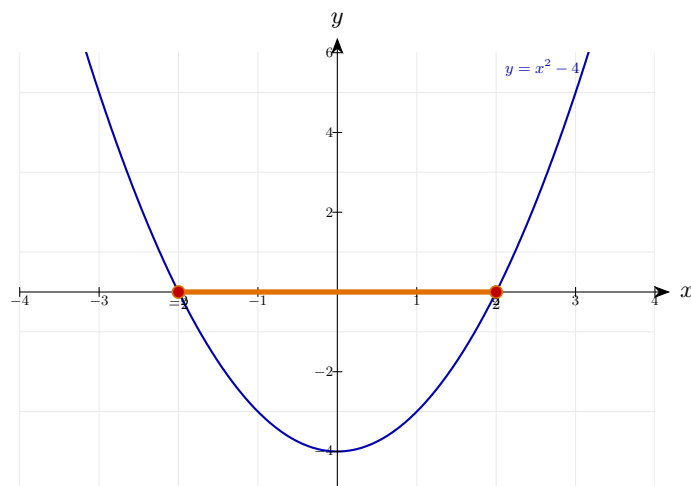
Step 3: State the solution.

$$-2 \leq x \leq 2$$

Closed interval $[-2, 2]$.

Final answer: $-2 \leq x \leq 2$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 14 (ise4-014)

Difficulty: Foundation / Marks: 3

Solve $x^2 - 8x + 15 \geq 0$.

Worked solution.

Step 1: Factorise.

$$x^2 - 8x + 15 = (x - 3)(x - 5)$$

Two numbers multiplying to 15 and summing to -8 are -3 and -5. Roots: $x = 3$, $x = 5$.

Step 2: Identify where the parabola is non-negative.

$$a = 1 > 0 \implies \geq 0 \text{ on } (-\infty, 3] \cup [5, \infty)$$

A U-shaped parabola is at or above the x -axis outside (and at) its roots.

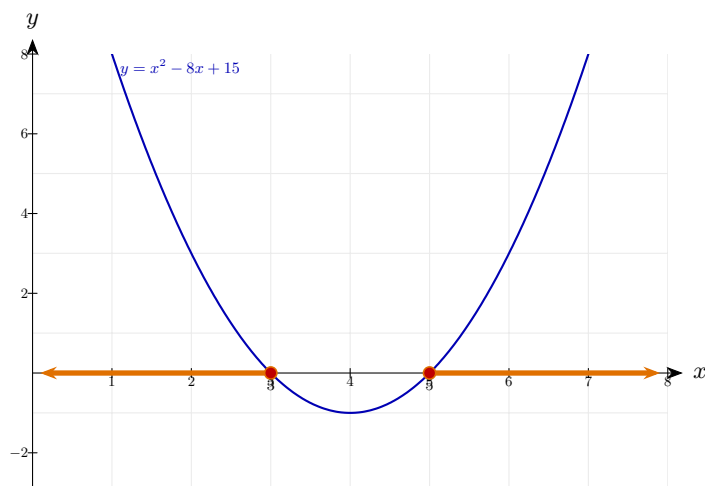
Step 3: State the solution.

$$x \leq 3 \text{ or } x \geq 5$$

Inclusive endpoints because of $\geq$.

Final answer: $x \leq 3$ or $x \geq 5$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 15 (ise4-015)

Difficulty: Foundation / Marks: 3

Solve $x^2 - 6x + 9 \geq 0$.

Worked solution.

Step 1: Recognise the perfect square.

$$x^2 - 6x + 9 = (x - 3)^2$$

Discriminant $= (-6)^2 - 4(1)(9) = 0$, confirming a repeated root at $x = 3$.

Step 2: Use the property that any real square is non-negative.

$$(x - 3)^2 \geq 0 \text{ for all } x \in \mathbb{R}$$

A squared real number can never be negative; it equals zero exactly at $x = 3$ and is positive everywhere else.

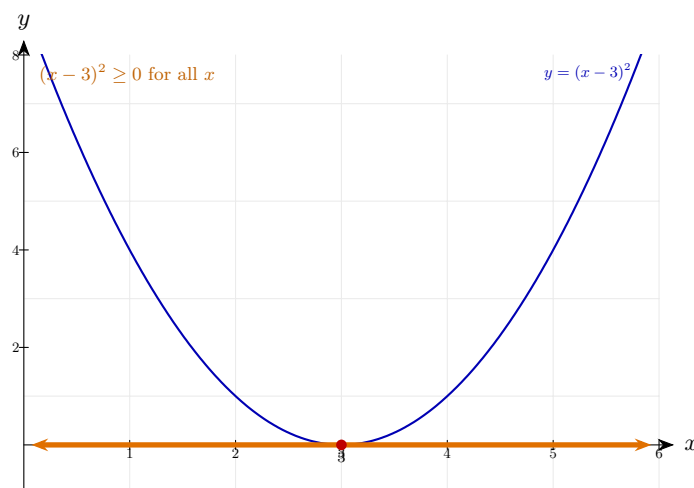
Step 3: State the solution.

$$x \in \mathbb{R}$$

The inequality is satisfied for every real number, including $x = 3$ where the expression equals 0.

Final answer: $x \in \mathbb{R}$ (all real numbers)

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 16 (ise4-016)

Difficulty: Foundation / Marks: 3

Solve $x^2 - 4x + 4 > 0$.

Worked solution.

Step 1: Recognise the perfect square.

$$x^2 - 4x + 4 = (x - 2)^2$$

Repeated root at $x = 2$.

Step 2: Use the property of real squares.

$$(x - 2)^2 \geq 0 \text{ for all } x, \text{ with equality only at } x = 2$$

The expression is strictly positive everywhere except at $x = 2$, where it is zero.

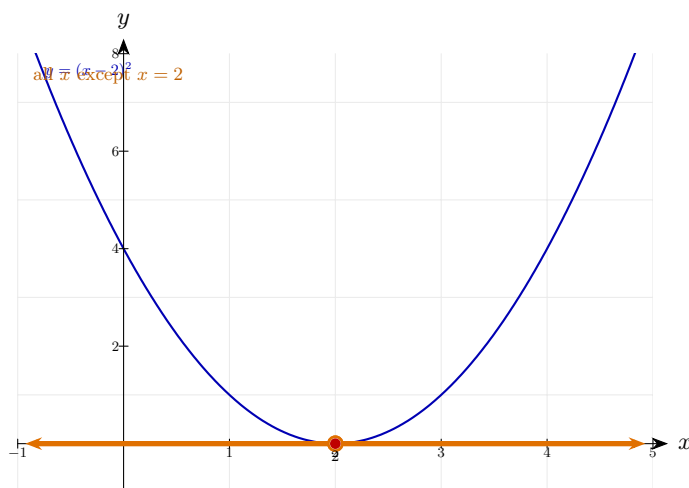
Step 3: Exclude the single point at which equality holds.

$$x \in \mathbb{R}, x \neq 2$$

Because the inequality is strict, we must exclude the value $x = 2$ which gives 0, not > 0 .

Final answer: $x \in \mathbb{R}, x \neq 2$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 17 (ise4-017)

Difficulty: Foundation / Marks: 3

Solve $x^2 + 10x + 25 < 0$.

Worked solution.

Step 1: Factorise as a perfect square.

$$x^2 + 10x + 25 = (x + 5)^2$$

Repeated root at $x = -5$.

Step 2: Apply the non-negativity of real squares.

$$(x + 5)^2 \geq 0 \text{ for all real } x$$

A real square is never negative. So $(x + 5)^2 < 0$ is impossible.

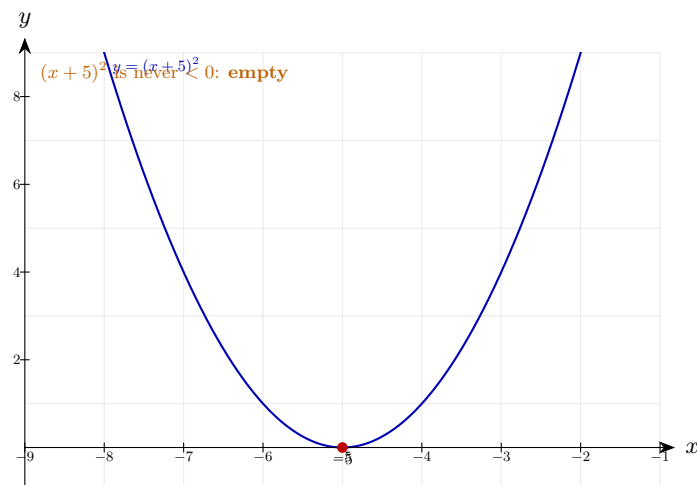
Step 3: State the conclusion.

no real solutions

The solution set is empty: there is no real x for which the expression is strictly less than zero.

Final answer: No real solutions (the solution set is empty)

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 18 (ise4-018)

Difficulty: Foundation / Marks: 3

Solve $x^2 + 4 > 0$.

Worked solution.

Step 1: Compute the discriminant.

$$\Delta = 0^2 - 4(1)(4) = -16 < 0$$

The discriminant is negative, so the quadratic has no real roots and the parabola never crosses the x -axis.

Step 2: Determine the constant sign of the quadratic.

$$a = 1 > 0 \implies x^2 + 4 > 0 \text{ for all } x$$

When $\Delta < 0$ the parabola lies entirely on one side of the x -axis, and its sign matches the sign of a . Here $a > 0$, so the expression is strictly positive everywhere.

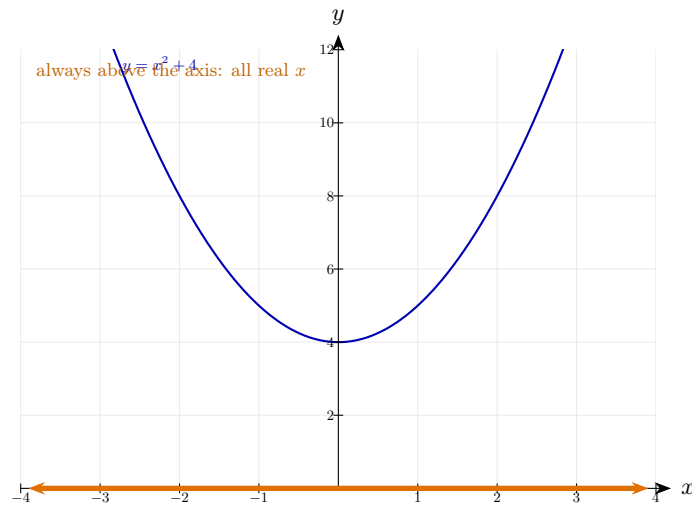
Step 3: State the solution.

$$x \in \mathbb{R}$$

The inequality holds for every real number.

Final answer: $x \in \mathbb{R}$ (all real numbers)

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 19 (ise4-019)

Difficulty: Foundation / Marks: 4

Solve $x^2 - 2x + 5 < 0$.

Worked solution.

Step 1: Compute the discriminant.

$$\Delta = (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$$

No real roots — the parabola never crosses the x -axis.

Step 2: Complete the square to confirm the sign.

$$x^2 - 2x + 5 = (x - 1)^2 + 4$$

Completing the square reveals that the minimum value of the expression is 4 (attained at $x = 1$). The expression is always at least 4, hence strictly positive.

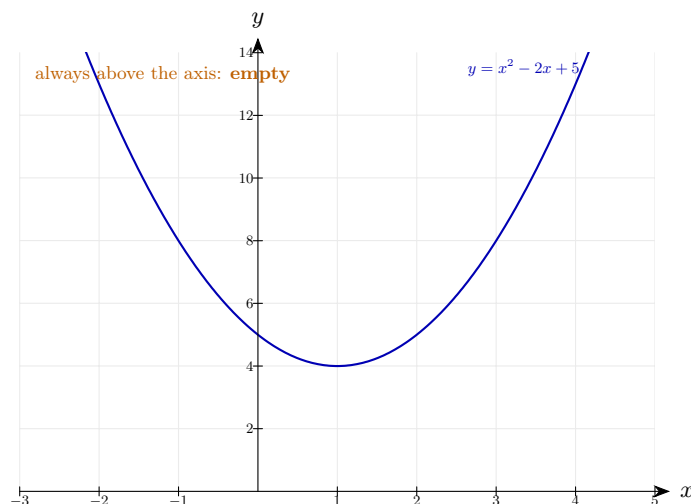
Step 3: Conclude.

$$x^2 - 2x + 5 > 0 \text{ for all } x \implies \text{no real solutions}$$

Since the expression is always positive, it can never be less than 0. The solution set is empty.

Final answer: No real solutions (the solution set is empty)

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 20 (ise4-020)

Difficulty: Foundation / Marks: 4

Solve $-x^2 + 2x - 5 < 0$.

Worked solution.

Step 1: Compute the discriminant.

$$\Delta = 2^2 - 4(-1)(-5) = 4 - 20 = -16 < 0$$

No real roots.

Step 2: Use the sign-of- a rule.

$$a = -1 < 0 \implies -x^2 + 2x - 5 < 0 \text{ for all } x$$

When $\Delta < 0$ the parabola has constant sign equal to the sign of a . Here $a < 0$, so the expression is strictly negative everywhere.

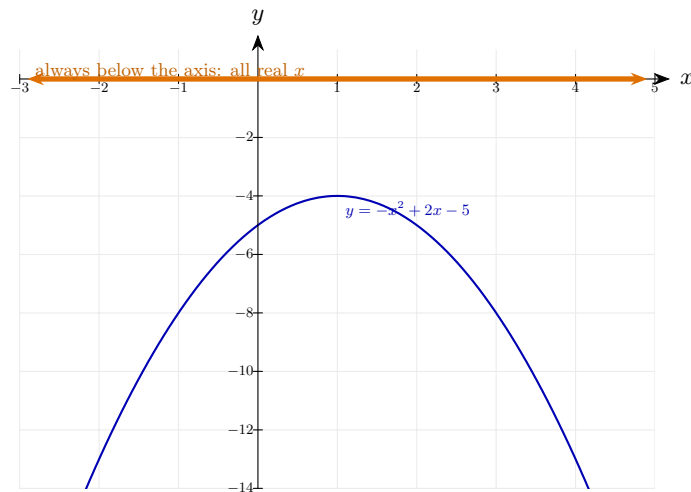
Step 3: State the solution.

$$x \in \mathbb{R}$$

The inequality is satisfied for every real x .

Final answer: $x \in \mathbb{R}$ (all real numbers)

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 21 (ise4-021)

Difficulty: Standard / Marks: 4

Solve $x^2 + 3x \geq 10$.

Worked solution.

Step 1: Rearrange to standard form (one side = 0).

$$x^2 + 3x - 10 \geq 0$$

You can only use the parabola sign-analysis once everything is on one side and compared to zero.

Step 2: Factorise.

$$x^2 + 3x - 10 = (x + 5)(x - 2)$$

Two numbers multiplying to -10 and summing to 3 are 5 and -2. Roots: $x = -5$, $x = 2$.

Step 3: Apply the U-shape rule with inclusive endpoints.

$$a > 0 \implies \geq 0 \text{ outside (and at) the roots}$$

A positive-leading-coefficient parabola is on or above the x -axis to the left of the smaller root and to the right of the larger root.

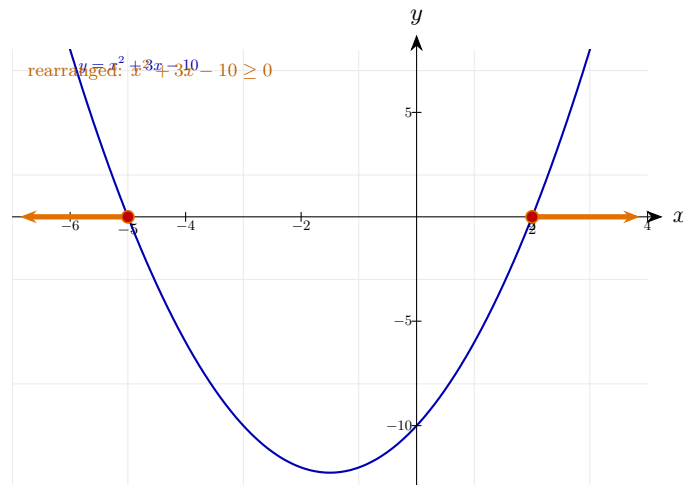
Step 4: State the answer.

$$x \leq -5 \text{ or } x \geq 2$$

Inclusive endpoints because the inequality is non-strict.

Final answer: $x \leq -5$ or $x \geq 2$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 22 (ise4-022)

Difficulty: Standard / Marks: 4

Solve $x^2 < 4x + 5$.

Worked solution.

Step 1: Bring all terms to one side.

$$x^2 - 4x - 5 < 0$$

Subtract $4x + 5$ from both sides so the right-hand side is zero.

Step 2: Factorise.

$$x^2 - 4x - 5 = (x - 5)(x + 1)$$

Roots: $x = -1$, $x = 5$.

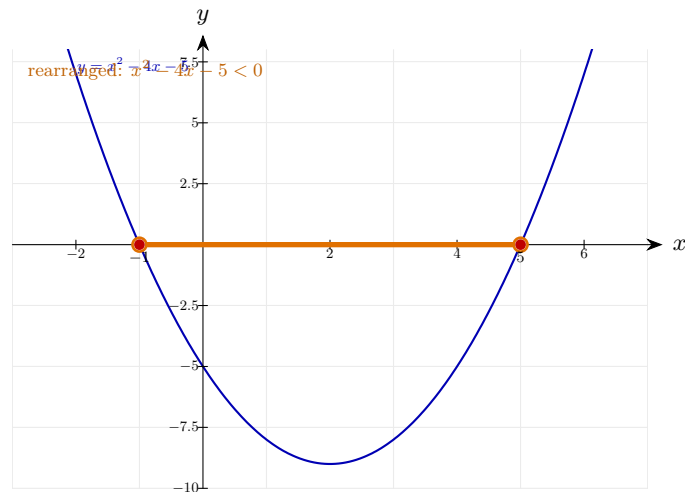
Step 3: Apply the U-shape rule.

$$(x - 5)(x + 1) < 0 \implies -1 < x < 5$$

Below zero between the roots for a U-shaped parabola.

Final answer: $-1 < x < 5$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 23 (ise4-023)

Difficulty: Standard / Marks: 4

Solve $2x^2 + 5 \geq 11x$.

Worked solution.

Step 1: Rearrange to standard form.

$$2x^2 - 11x + 5 \geq 0$$

Subtract $11x$ from both sides.

Step 2: Factorise the non-monic quadratic.

$$2x^2 - 11x + 5 = (2x - 1)(x - 5)$$

Look for factors of $2 \times 5 = 10$ that sum to -11 : these are -1 and -10 . Split the middle term: $2x^2 - x - 10x + 5 = x(2x - 1) - 5(2x - 1) = (2x - 1)(x - 5)$. Roots: $x = \frac{1}{2}$, $x = 5$.

Step 3: Apply the U-shape rule.

$$a = 2 > 0 \implies \geq 0 \text{ outside (and at) the roots}$$

Still U-shaped because the leading coefficient is positive — only its steepness changes.

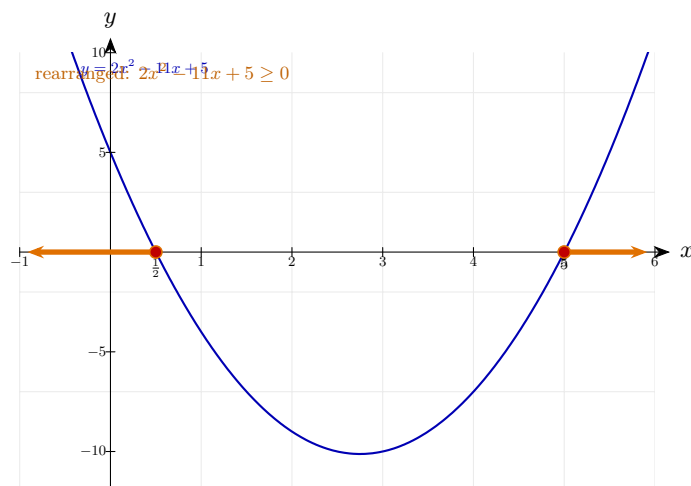
Step 4: State the solution.

$$x \leq \frac{1}{2} \text{ or } x \geq 5$$

Inclusive endpoints.

Final answer: $x \leq \frac{1}{2} \text{ or } x \geq 5$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 24 (ise4-024)

Difficulty: Standard / Marks: 4

Solve $x(x - 1) \leq 6$.

Worked solution.

Step 1: Expand and bring to standard form.

$$x^2 - x \leq 6 \iff x^2 - x - 6 \leq 0$$

Resist the temptation to read roots straight from $x(x - 1) = 6$: the factor product equals 6, not 0, so this is not a difference-equals-zero situation. Expand first.

Step 2: Factorise.

$$x^2 - x - 6 = (x - 3)(x + 2)$$

Roots: $x = -2$, $x = 3$.

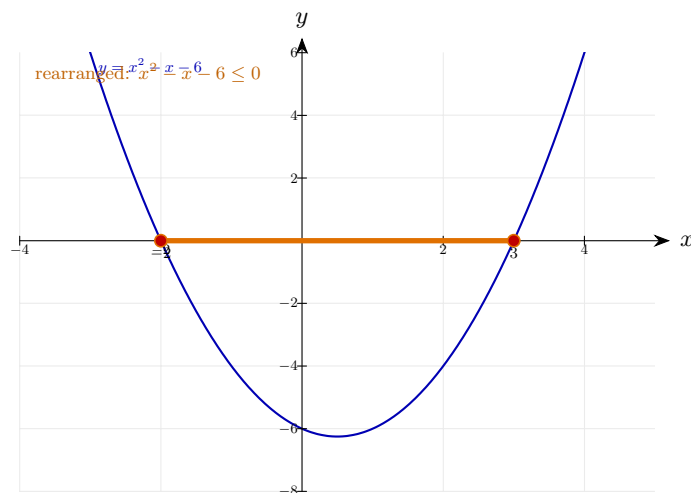
Step 3: Apply the U-shape rule with inclusive endpoints.

$$(x - 3)(x + 2) \leq 0 \implies -2 \leq x \leq 3$$

Closed interval because of $\leq$.

Final answer: $-2 \leq x \leq 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 25 (ise4-025)

Difficulty: Standard / Marks: 4

Solve $2x^2 + 5x - 3 < 0$.

Worked solution.

Step 1: Factorise the non-monic quadratic.

$$2x^2 + 5x - 3 = (2x - 1)(x + 3)$$

Two numbers multiplying to $2 \times (-3) = -6$ and summing to 5 are 6 and -1. Split: $2x^2 + 6x - x - 3 = 2x(x + 3) - (x + 3) = (2x - 1)(x + 3)$. Roots: $x = -3$, $x = \frac{1}{2}$.

Step 2: Apply the U-shape rule.

$$a = 2 > 0 \implies < 0 \text{ between the roots}$$

Positive leading coefficient gives a U-shape; the expression dips below zero strictly between the roots.

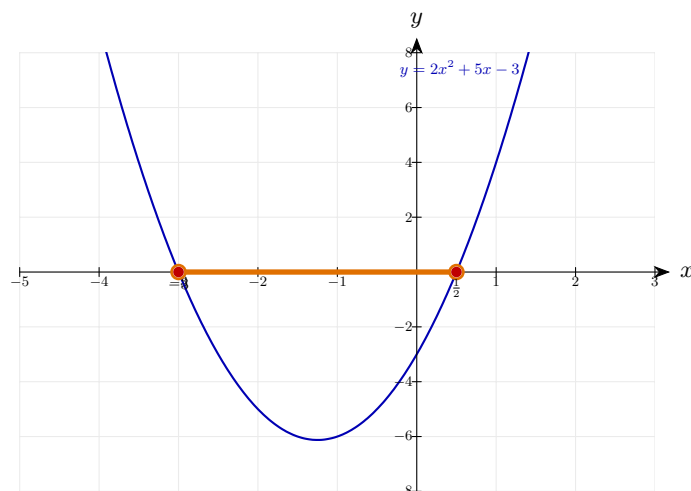
Step 3: State the solution.

$$-3 < x < \frac{1}{2}$$

Open interval $(-3, \frac{1}{2})$.

Final answer: $-3 < x < \frac{1}{2}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 26 (ise4-026)

Difficulty: Standard / Marks: 4

Solve $3x^2 - 7x + 2 \geq 0$.

Worked solution.

Step 1: Factorise.

$$3x^2 - 7x + 2 = (3x - 1)(x - 2)$$

Two numbers multiplying to $3 \times 2 = 6$ and summing to -7 are -1 and -6 . Split: $3x^2 - x - 6x + 2 = x(3x - 1) - 2(3x - 1) = (3x - 1)(x - 2)$. Roots: $x = \frac{1}{3}$, $x = 2$.

Step 2: Apply the U-shape rule with inclusive endpoints.

$$a = 3 > 0 \implies \geq 0 \text{ outside (and at) the roots}$$

U-shape because the leading coefficient is positive.

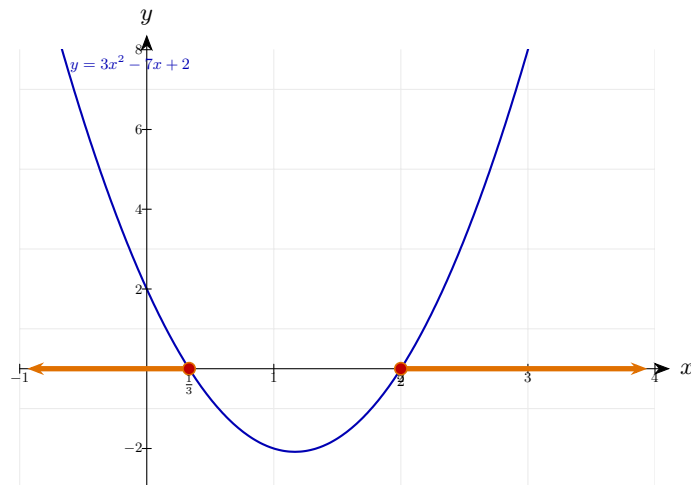
Step 3: State the solution.

$$x \leq \frac{1}{3} \text{ or } x \geq 2$$

Inclusive endpoints because of $\geq$.

Final answer: $x \leq \frac{1}{3}$ or $x \geq 2$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 27 (ise4-027)

Difficulty: Standard / Marks: 4

Solve $6x^2 - x - 1 \leq 0$.

Worked solution.

Step 1: Factorise.

$$6x^2 - x - 1 = (3x + 1)(2x - 1)$$

Two numbers multiplying to $6 \times (-1) = -6$ and summing to -1 are -3 and 2 . Split: $6x^2 - 3x + 2x - 1 = 3x(2x - 1) + (2x - 1) = (3x + 1)(2x - 1)$. Roots: $x = -\frac{1}{3}$, $x = \frac{1}{2}$.

Step 2: Apply the U-shape rule.

$$a = 6 > 0 \implies \leq 0 \text{ on the closed interval between the roots}$$

U-shape; below or on the x -axis from the smaller root to the larger one.

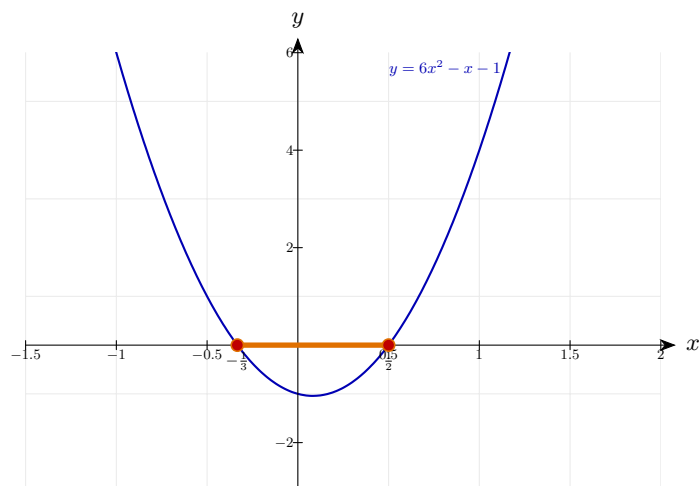
Step 3: State the solution.

$$-\frac{1}{3} \leq x \leq \frac{1}{2}$$

Closed interval.

Final answer: $-\frac{1}{3} \leq x \leq \frac{1}{2}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 28 (ise4-028)

Difficulty: Standard / Marks: 4

Solve $4x^2 - 12x + 9 > 0$.

Worked solution.

Step 1: Spot the perfect square.

$$4x^2 - 12x + 9 = (2x - 3)^2$$

Discriminant $= (-12)^2 - 4(4)(9) = 0$, confirming a repeated root at $x = \frac{3}{2}$.

Step 2: Apply the non-negativity of real squares.

$$(2x - 3)^2 \geq 0 \text{ for all } x, \text{ with equality only at } x = \frac{3}{2}$$

Squares of real numbers cannot be negative, so the expression is at least 0 everywhere and exactly 0 only at $x = \frac{3}{2}$.

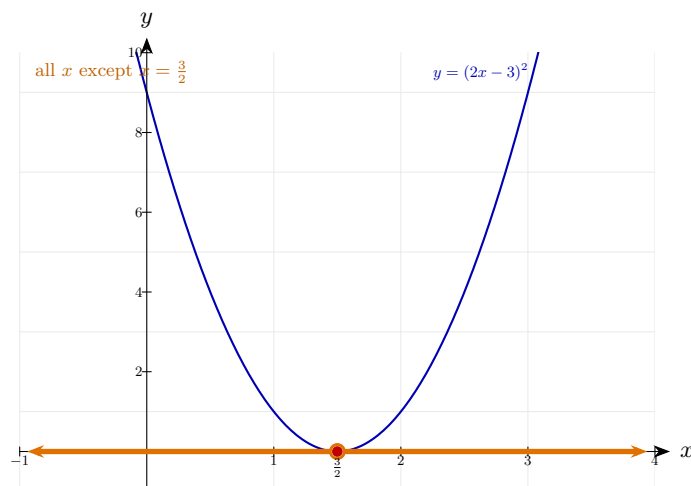
Step 3: Exclude the point where equality holds.

$$x \in \mathbb{R}, x \neq \frac{3}{2}$$

Because the inequality is strict, the single point at which the expression equals zero is excluded.

Final answer: $x \in \mathbb{R}, x \neq \frac{3}{2}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 29 (ise4-029)

Difficulty: Standard / Marks: 5

Solve $x^2 + 4x - 1 < 0$. Give your answer in exact (surd) form.

Worked solution.

Step 1: Find the roots using the quadratic formula.

$$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-1)}}{2(1)} = \frac{-4 \pm \sqrt{20}}{2}$$

The expression does not factorise over the integers, so use $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Discriminant = $16 + 4 = 20 > 0$, so there are two real roots.

Step 2: Simplify the surd.

$$\sqrt{20} = 2\sqrt{5} \implies x = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}$$

Roots: $x = -2 - \sqrt{5}$ and $x = -2 + \sqrt{5}$.

Step 3: Apply the U-shape rule.

$$a = 1 > 0 \implies < 0 \text{ strictly between the roots}$$

For a U-shaped parabola the expression is negative on the open interval between the two roots.

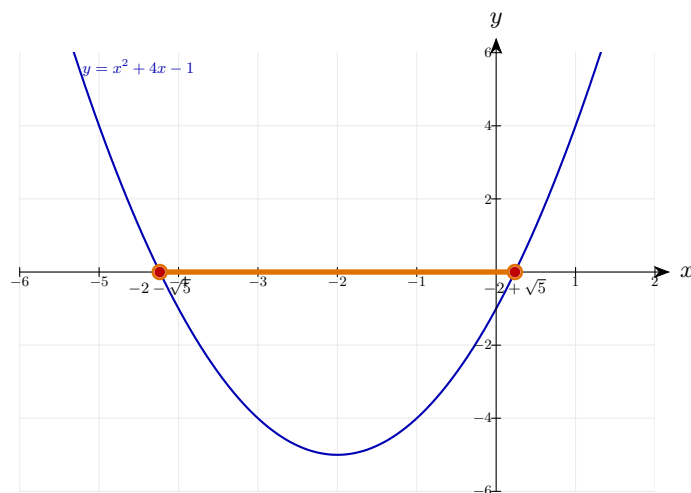
Step 4: State the solution.

$$-2 - \sqrt{5} < x < -2 + \sqrt{5}$$

Open interval; the endpoints are excluded.

Final answer: $-2 - \sqrt{5} < x < -2 + \sqrt{5}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 30 (ise4-030)

Difficulty: Standard / Marks: 5

Solve $x^2 - 6x + 4 > 0$. Give your answer in exact form.

Worked solution.

Step 1: Find the roots using the quadratic formula.

$$x = \frac{6 \pm \sqrt{36 - 16}}{2} = \frac{6 \pm \sqrt{20}}{2}$$

Discriminant $= 20 > 0$ so there are two distinct real roots.

Step 2: Simplify the surd.

$$x = \frac{6 \pm 2\sqrt{5}}{2} = 3 \pm \sqrt{5}$$

Roots: $x = 3 - \sqrt{5}$ and $x = 3 + \sqrt{5}$.

Step 3: Apply the U-shape rule.

$$a = 1 > 0 \implies > 0 \text{ strictly outside the roots}$$

A U-shape is above the x -axis to the left of the smaller root and to the right of the larger root.

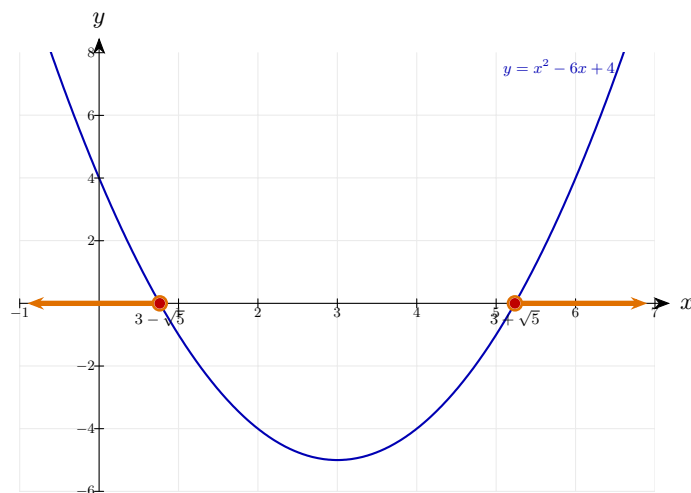
Step 4: State the solution.

$$x < 3 - \sqrt{5} \text{ or } x > 3 + \sqrt{5}$$

Strict inequality so the endpoints are excluded.

Final answer: $x < 3 - \sqrt{5}$ or $x > 3 + \sqrt{5}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 31 (ise4-031)

Difficulty: Standard / Marks: 5

Solve $2x^2 - 4x - 1 \leq 0$. Give your answer in exact form.

Worked solution.

Step 1: Apply the quadratic formula.

$$x = \frac{4 \pm \sqrt{16 - 4(2)(-1)}}{2(2)} = \frac{4 \pm \sqrt{24}}{4}$$

Discriminant $= 16 + 8 = 24 > 0$. Note the careful use of $2a = 4$ in the denominator since $a = 2$.

Step 2: Simplify.

$$\sqrt{24} = 2\sqrt{6} \implies x = \frac{4 \pm 2\sqrt{6}}{4} = \frac{2 \pm \sqrt{6}}{2}$$

Cancel a common factor of 2 from numerator and denominator.

Step 3: Apply the U-shape rule with inclusive endpoints.

$$a = 2 > 0 \implies \leq 0 \text{ on the closed interval between the roots}$$

U-shape; below or on the x -axis between (and at) its two roots.

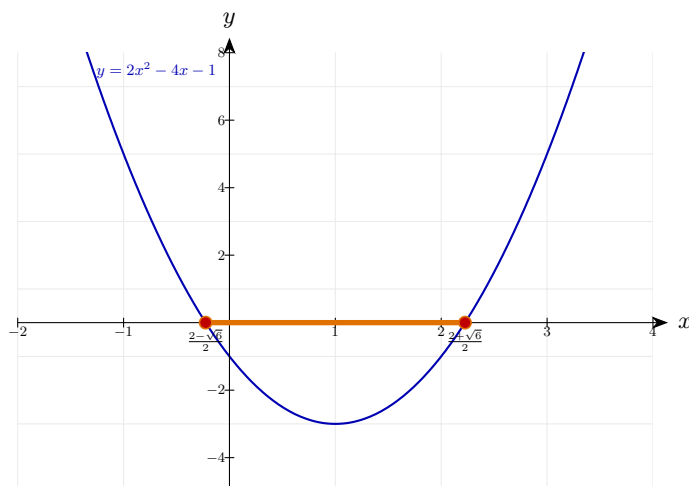
Step 4: State the solution.

$$\frac{2 - \sqrt{6}}{2} \leq x \leq \frac{2 + \sqrt{6}}{2}$$

Closed interval; endpoints are included because the inequality is non-strict.

Final answer: $\frac{2 - \sqrt{6}}{2} \leq x \leq \frac{2 + \sqrt{6}}{2}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 32 (ise4-032)

Difficulty: Standard / Marks: 5

Solve $x^2 + 2x - 4 \geq 0$. Give your answer in exact form.

Worked solution.

Step 1: Apply the quadratic formula.

$$x = \frac{-2 \pm \sqrt{4 - 4(1)(-4)}}{2} = \frac{-2 \pm \sqrt{20}}{2}$$

Discriminant $= 4 + 16 = 20$.

Step 2: Simplify.

$$x = \frac{-2 \pm 2\sqrt{5}}{2} = -1 \pm \sqrt{5}$$

Cancel a factor of 2 in numerator and denominator.

Step 3: Apply the U-shape rule with inclusive endpoints.

$$a = 1 > 0 \implies \geq 0 \text{ outside (and at) the roots}$$

Above or on the x -axis outside the two roots.

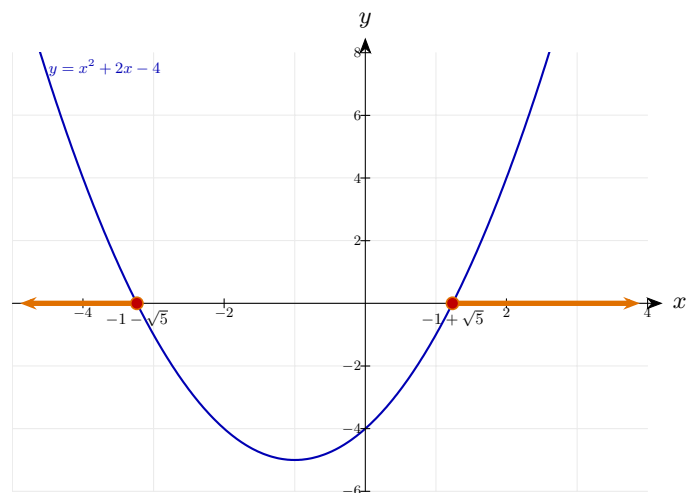
Step 4: State the solution.

$$x \leq -1 - \sqrt{5} \text{ or } x \geq -1 + \sqrt{5}$$

Inclusive endpoints because of $\geq$.

Final answer: $x \leq -1 - \sqrt{5}$ or $x \geq -1 + \sqrt{5}$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 33 (ise4-033)

Difficulty: Standard / Marks: 5

Find the set of values of x that satisfy $-2 < x^2 - 3 < 6$.

Worked solution.

Step 1: Split into two simultaneous inequalities.

$$(A) \ x^2 - 3 > -2 \quad (B) \ x^2 - 3 < 6$$

A double inequality is shorthand for the intersection of the two single inequalities — both must hold simultaneously.

Step 2: Solve (A).

$$x^2 > 1 \implies x < -1 \text{ or } x > 1$$

A common trap is to write $x > 1$ only and forget the negative branch. The inequality $x^2 > 1$ really says $|x| > 1$, which gives two intervals.

Step 3: Solve (B).

$$x^2 < 9 \implies -3 < x < 3$$

$x^2 < 9$ means $|x| < 3$, giving a single open interval.

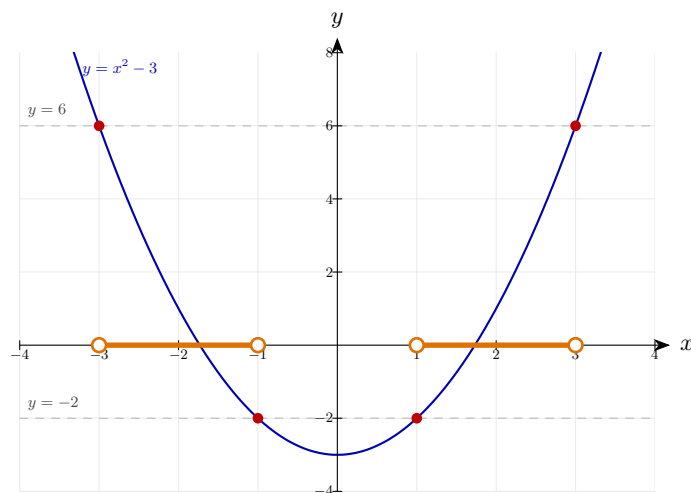
Step 4: Intersect the two solution sets.

$$(x < -1 \text{ or } x > 1) \cap (-3 < x < 3) = -3 < x < -1 \text{ or } 1 < x < 3$$

Restrict each branch of (A) to the interval $(-3, 3)$ given by (B). The left branch $x < -1$ becomes $-3 < x < -1$; the right branch $x > 1$ becomes $1 < x < 3$.

Final answer: $-3 < x < -1$ or $1 < x < 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 34 (ise4-034)

Difficulty: Standard / Marks: 5

Find the set of values of x that satisfy $0 \leq x^2 - 4 \leq 5$.

Worked solution.

Step 1: Split the double inequality.

$$(A) \ x^2 - 4 \geq 0 \quad (B) \ x^2 - 4 \leq 5$$

Both inequalities must hold simultaneously.

Step 2: Solve (A).

$$x^2 \geq 4 \implies x \leq -2 \text{ or } x \geq 2$$

Equivalent to $|x| \geq 2$.

Step 3: Solve (B).

$$x^2 \leq 9 \implies -3 \leq x \leq 3$$

Equivalent to $|x| \leq 3$.

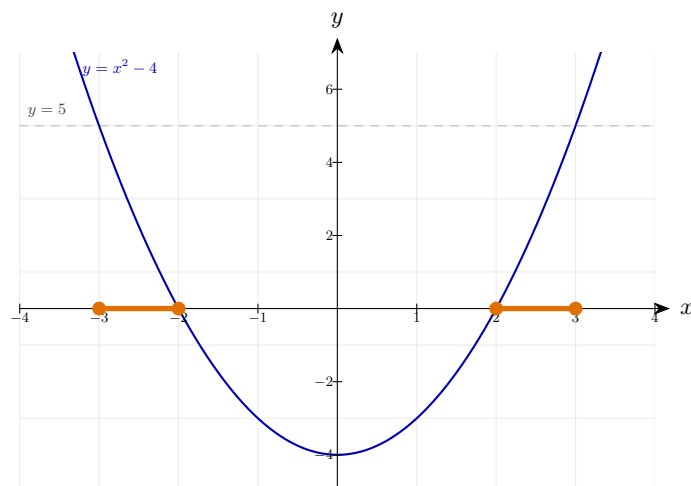
Step 4: Intersect.

$$-3 \leq x \leq -2 \text{ or } 2 \leq x \leq 3$$

Take the parts of $[-3, 3]$ that also satisfy $|x| \geq 2$.

Final answer: $-3 \leq x \leq -2$ or $2 \leq x \leq 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 35 (ise4-035)

Difficulty: Standard / Marks: 6

Find the set of values of x that satisfy $-1 < x^2 - 2x < 8$.

Worked solution.

Step 1: Split into two inequalities.

$$(A) \ x^2 - 2x > -1 \quad (B) \ x^2 - 2x < 8$$

Both must hold.

Step 2: Solve (A).

$$x^2 - 2x + 1 > 0 \iff (x - 1)^2 > 0$$

Rearrange and recognise the perfect square. A real square is strictly positive everywhere except at $x = 1$. So (A) holds for all $x \neq 1$.

Step 3: Solve (B).

$$x^2 - 2x - 8 < 0 \iff (x - 4)(x + 2) < 0 \implies -2 < x < 4$$

Standard factor-and-sign analysis: U-shape, negative between the roots.

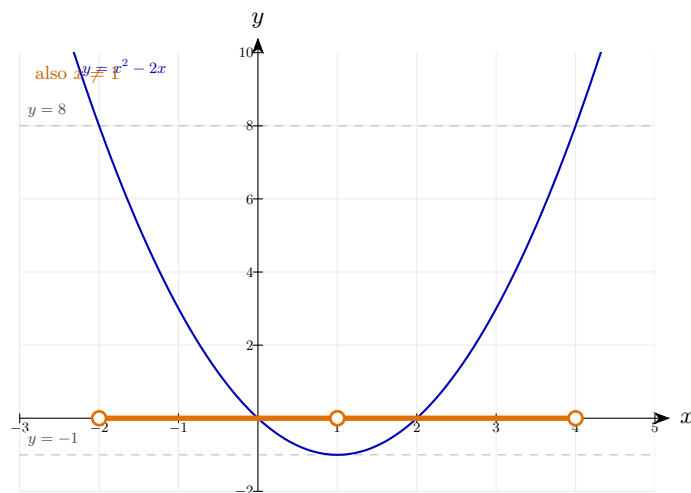
Step 4: Intersect.

$$(-2 < x < 4) \cap (x \neq 1) = -2 < x < 4, \ x \neq 1$$

Remove the single point $x = 1$ from the open interval $(-2, 4)$, because at $x = 1$ inequality (A) fails (the left bound -1 is not strictly less than $x^2 - 2x = -1$).

Final answer: $-2 < x < 4, \ x \neq 1$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 36 (ise4-036)

Difficulty: Standard / Marks: 6

Find the set of values of x that satisfy $3 \leq x^2 + 2x \leq 8$.

Worked solution.

Step 1: Split the double inequality.

$$(A) \ x^2 + 2x \geq 3 \quad (B) \ x^2 + 2x \leq 8$$

Both must hold simultaneously.

Step 2: Solve (A).

$$x^2 + 2x - 3 \geq 0 \iff (x + 3)(x - 1) \geq 0 \implies x \leq -3 \text{ or } x \geq 1$$

U-shape, non-negative outside (and at) the roots.

Step 3: Solve (B).

$$x^2 + 2x - 8 \leq 0 \iff (x + 4)(x - 2) \leq 0 \implies -4 \leq x \leq 2$$

U-shape, non-positive between (and at) the roots.

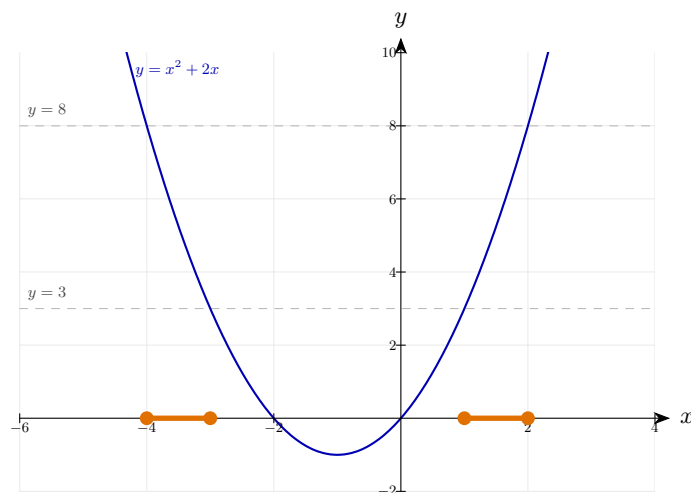
Step 4: Intersect.

$$-4 \leq x \leq -3 \text{ or } 1 \leq x \leq 2$$

Restrict each branch of (A) to $[-4, 2]$: the branch $x \leq -3$ becomes $[-4, -3]$; the branch $x \geq 1$ becomes $[1, 2]$.

Final answer: $-4 \leq x \leq -3$ or $1 \leq x \leq 2$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 37 (ise4-037)

Difficulty: Standard / Marks: 5

Find the set of values of x for which both $x^2 - 4x + 3 \leq 0$ and $x > 1$ hold.

Worked solution.

Step 1: Solve the quadratic inequality.

$$x^2 - 4x + 3 = (x - 1)(x - 3) \leq 0 \implies 1 \leq x \leq 3$$

U-shape, non-positive between (and at) the roots $x = 1$ and $x = 3$.

Step 2: Combine with the linear condition $x > 1$.

$$[1, 3] \cap (1, \infty) = (1, 3]$$

The intersection drops the left endpoint $x = 1$ (excluded by the strict linear inequality) but keeps $x = 3$ (included by the non-strict quadratic inequality).

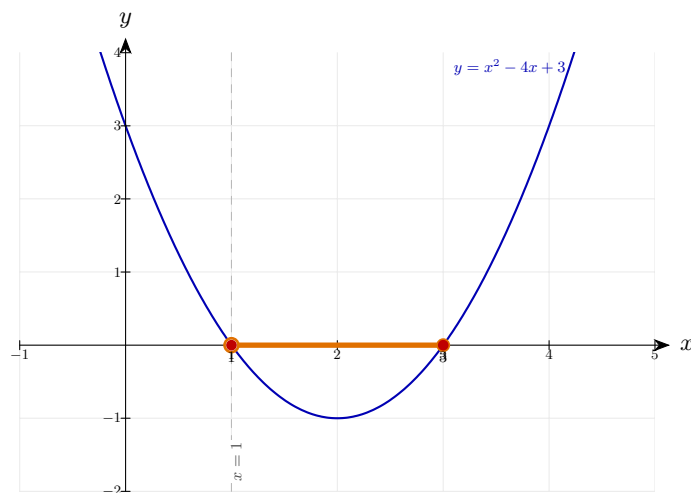
Step 3: State the solution.

$$1 < x \leq 3$$

A half-open interval — pay attention to mixed strictness.

Final answer: $1 < x \leq 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 38 (ise4-038)

Difficulty: Standard / Marks: 5

Find the set of values of x for which both $x^2 - 9 \geq 0$ and $2x + 1 < 9$ hold.

Worked solution.

Step 1: Solve the quadratic inequality.

$$x^2 - 9 \geq 0 \iff (x - 3)(x + 3) \geq 0 \implies x \leq -3 \text{ or } x \geq 3$$

U-shape, non-negative outside (and at) the roots $x = \pm 3$.

Step 2: Solve the linear inequality.

$$2x + 1 < 9 \implies x < 4$$

Subtract 1, divide by 2 (positive, no flip).

Step 3: Intersect the two solution sets.

$$(x \leq -3 \text{ or } x \geq 3) \cap (x < 4)$$

Restrict each branch to $x < 4$.

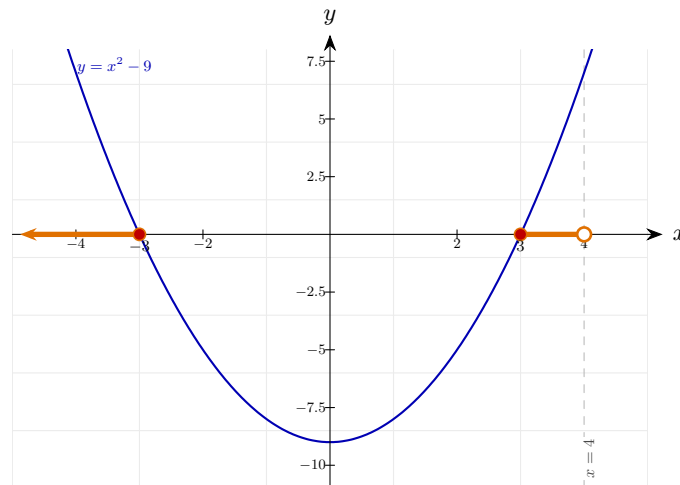
Step 4: State the answer.

$$x \leq -3 \text{ or } 3 \leq x < 4$$

The left branch $x \leq -3$ is already inside $x < 4$, so unchanged. The right branch $x \geq 3$ becomes the half-open interval $[3, 4)$.

Final answer: $x \leq -3$ or $3 \leq x < 4$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 39 (ise4-039)

Difficulty: Standard / Marks: 5

Find the set of values of x for which both $2x^2 - 5x - 3 < 0$ and $x \geq 0$ hold. Give your answer in interval notation.

Worked solution.

Step 1: Factorise the quadratic.

$$2x^2 - 5x - 3 = (2x + 1)(x - 3)$$

Two numbers multiplying to $2 \times (-3) = -6$ and summing to -5 are -6 and 1 . Split: $2x^2 - 6x + x - 3 = 2x(x - 3) + (x - 3) = (2x + 1)(x - 3)$. Roots: $x = -\frac{1}{2}$, $x = 3$.

Step 2: Solve the quadratic inequality.

$$(2x + 1)(x - 3) < 0 \implies -\frac{1}{2} < x < 3$$

U-shape ($a = 2 > 0$), strictly negative between the roots.

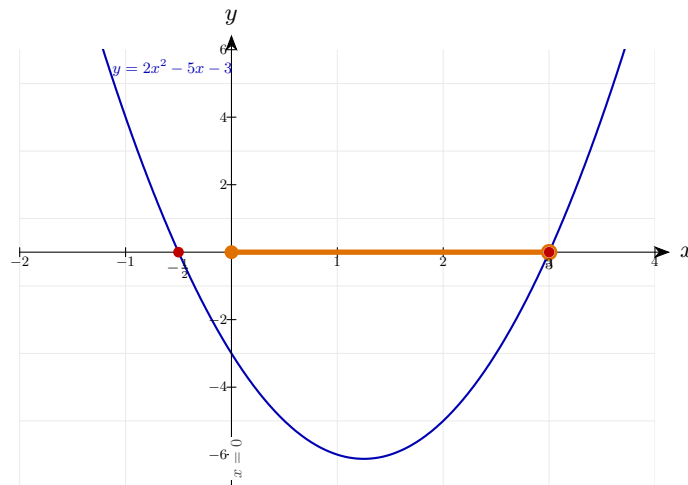
Step 3: Intersect with $x \geq 0$.

$$(-\frac{1}{2}, 3) \cap [0, \infty) = [0, 3)$$

The left endpoint becomes 0 (included via $x \geq 0$); the right endpoint 3 is excluded (strict from the quadratic).

Final answer: $x \in [0, 3)$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 40 (ise4-040)

Difficulty: Standard / Marks: 5

Find the set of values of x for which both $x^2 + x - 6 > 0$ and $3 - x \geq 0$ hold.

Worked solution.

Step 1: Solve the quadratic inequality.

$$x^2 + x - 6 = (x + 3)(x - 2) > 0 \implies x < -3 \text{ or } x > 2$$

U-shape, strictly positive outside the roots.

Step 2: Solve the linear inequality.

$$3 - x \geq 0 \implies x \leq 3$$

Rearrange — add x to both sides, no flip needed.

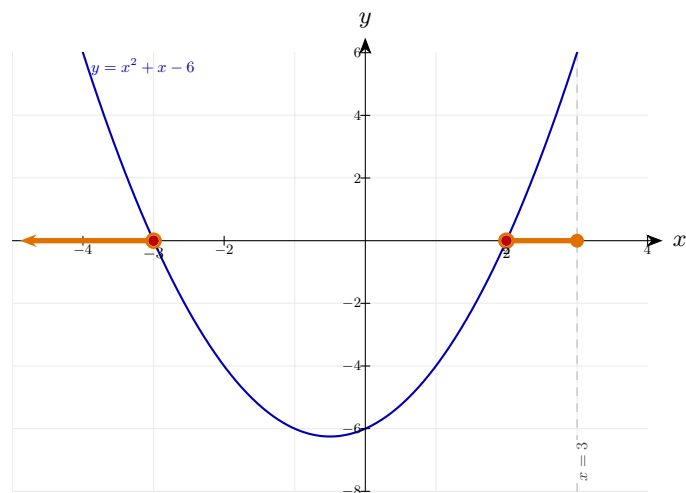
Step 3: Intersect.

$$(x < -3 \text{ or } x > 2) \cap (x \leq 3) = x < -3 \text{ or } 2 < x \leq 3$$

The left branch is unchanged; the right branch $x > 2$ is capped at 3 (inclusive), giving $(2, 3]$.

Final answer: $x < -3$ or $2 < x \leq 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 41 (ise4-041)

Difficulty: Challenge / Marks: 5

Find the set of values of the constant k for which the inequality $x^2 + kx + 9 > 0$ holds for all real values of x .

Worked solution.

Step 1: Identify the condition for "positive for all real x ".

$$a = 1 > 0; \text{ need parabola to never touch / cross the } x\text{-axis}$$

A U-shaped parabola is strictly above the x -axis everywhere if and only if it has no real roots. This translates to the discriminant being strictly negative.

Step 2: Set up the discriminant condition.

$$b^2 - 4ac < 0 \implies k^2 - 4(1)(9) < 0 \implies k^2 - 36 < 0$$

Here the variable in the original quadratic is x , and k plays the role of b . The discriminant in x depends on k .

Step 3: Solve the resulting quadratic inequality in k .

$$k^2 - 36 < 0 \iff (k - 6)(k + 6) < 0 \implies -6 < k < 6$$

A second quadratic inequality appears, now in the variable k . Same technique as before: U-shape, negative between the roots.

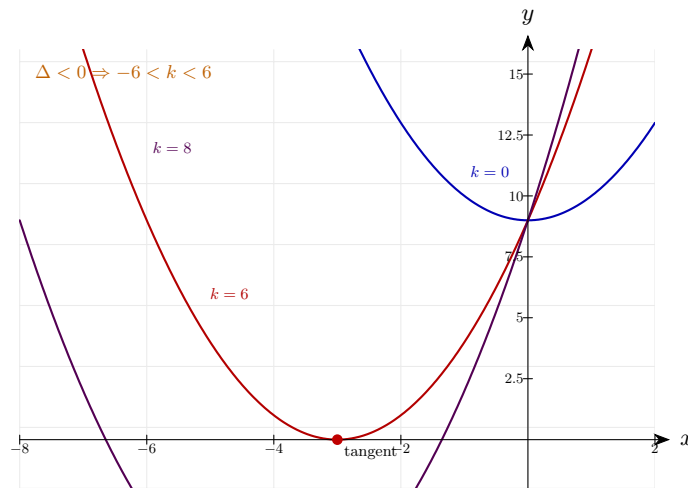
Step 4: State the answer.

$$-6 < k < 6$$

For these values of k the original quadratic in x is positive for every real x .

Final answer: $-6 < k < 6$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 42 (ise4-042)

Difficulty: Challenge / Marks: 5

Find the set of values of k for which the equation $2x^2 + kx + 8 = 0$ has no real roots.

Worked solution.

Step 1: State the condition for no real roots.

$$\Delta < 0$$

A quadratic equation has no real roots exactly when its discriminant is strictly negative.

Step 2: Form the discriminant.

$$\Delta = k^2 - 4(2)(8) = k^2 - 64$$

With $a = 2, b = k, c = 8$.

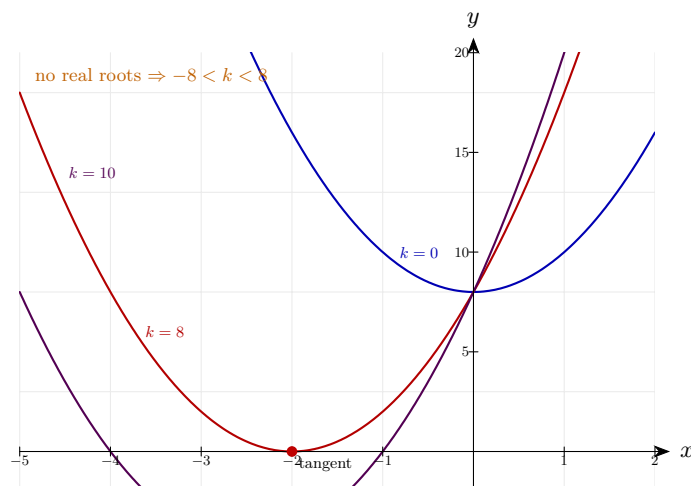
Step 3: Solve $\Delta < 0$.

$$k^2 - 64 < 0 \iff (k - 8)(k + 8) < 0 \implies -8 < k < 8$$

U-shape (in k); negative between the roots.

Final answer: $-8 < k < 8$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 43 (ise4-043)

Difficulty: Challenge / Marks: 6

Find the set of values of p for which the equation $x^2 + (p - 2)x + 4 = 0$ has two distinct real roots.

Worked solution.

Step 1: State the condition for two distinct real roots.

$$\Delta > 0$$

Two distinct real roots correspond to a strictly positive discriminant.

Step 2: Compute the discriminant.

$$\Delta = (p - 2)^2 - 4(1)(4) = (p - 2)^2 - 16$$

Here $b = p - 2$, so be careful to square the entire expression — students often write $p^2 - 4$ by mistake.

Step 3: Solve $(p - 2)^2 - 16 > 0$.

$$(p - 2)^2 > 16 \implies |p - 2| > 4 \implies p - 2 < -4 \text{ or } p - 2 > 4$$

Take the square root after isolating the squared term. The square-root step introduces $|\cdot|$ and hence two cases.

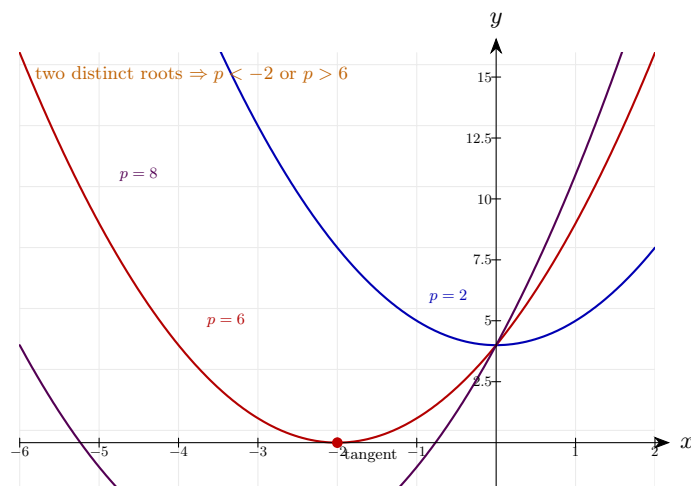
Step 4: Simplify each branch.

$$p < -2 \text{ or } p > 6$$

Add 2 to both sides of each case.

Final answer: $p < -2$ or $p > 6$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 44 (ise4-044)

Difficulty: Challenge / Marks: 4

Solve $(x - 1)^2 - 9 \leq 0$ without expanding the bracket.

Worked solution.

Step 1: Isolate the squared term.

$$(x - 1)^2 \leq 9$$

Add 9 to both sides. Treat $x - 1$ as a single variable.

Step 2: Take square roots, introducing $|\cdot|$.

$$|x - 1| \leq 3 \iff -3 \leq x - 1 \leq 3$$

For any non-negative k , $y^2 \leq k^2$ is equivalent to $|y| \leq k$, which is equivalent to $-k \leq y \leq k$.

Step 3: Solve the double inequality.

$$-2 \leq x \leq 4$$

Add 1 to each part of the chain.

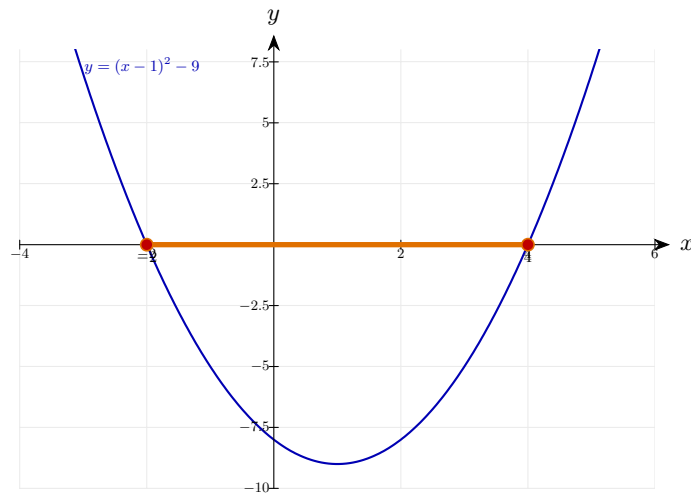
Step 4: Sanity check by expanding.

$$(x - 1)^2 - 9 = x^2 - 2x - 8 = (x - 4)(x + 2)$$

Roots $x = -2$ and $x = 4$ match — the answer is correct.

Final answer: $-2 \leq x \leq 4$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 45 (ise4-045)

Difficulty: Challenge / Marks: 4

Solve $(2x + 3)^2 > 25$.

Worked solution.

Step 1: Translate the squared inequality to an absolute-value statement.

$$(2x + 3)^2 > 25 \iff |2x + 3| > 5$$

For any $k > 0$, $y^2 > k^2$ is equivalent to $|y| > k$.

Step 2: Break into two cases.

$$2x + 3 < -5 \text{ or } 2x + 3 > 5$$

$|y| > 5$ splits into $y < -5$ or $y > 5$ — two disjoint pieces.

Step 3: Solve each linear inequality.

$$2x < -8 \Rightarrow x < -4 \quad \text{or} \quad 2x > 2 \Rightarrow x > 1$$

Subtract 3 then divide by 2 (positive — no flip).

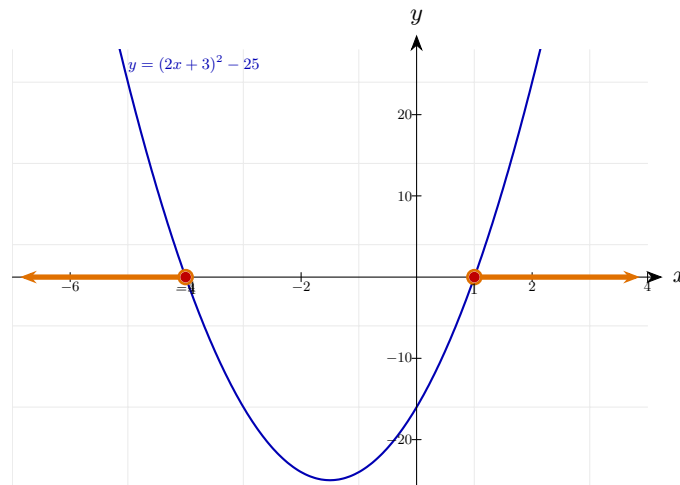
Step 4: State the solution.

$$x < -4 \text{ or } x > 1$$

Union of two open intervals.

Final answer: $x < -4$ or $x > 1$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 46 (ise4-046)

Difficulty: Challenge / Marks: 6

Solve $x^4 - 5x^2 + 4 \leq 0$. [Hint: substitute $u = x^2$.]

Worked solution.

Step 1: Substitute $u = x^2$.

$$u^2 - 5u + 4 \leq 0$$

A quadratic in u . Note that $u = x^2 \geq 0$ automatically, which we must remember when translating back.

Step 2: Factorise and solve in u .

$$(u - 1)(u - 4) \leq 0 \implies 1 \leq u \leq 4$$

U-shape; non-positive between (and at) the roots.

Step 3: Translate back to x .

$$1 \leq x^2 \leq 4$$

Replace u with x^2 . This is now a double-bounded inequality in x^2 .

Step 4: Solve the double inequality.

$$x^2 \geq 1 \text{ and } x^2 \leq 4$$

$$x^2 \geq 1 \Leftrightarrow |x| \geq 1; x^2 \leq 4 \Leftrightarrow |x| \leq 2. \text{ Together: } 1 \leq |x| \leq 2.$$

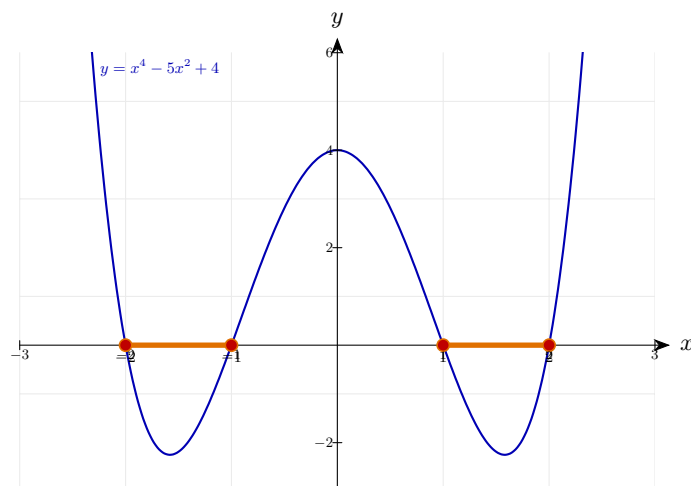
Step 5: Convert the absolute-value statement to intervals.

$$-2 \leq x \leq -1 \text{ or } 1 \leq x \leq 2$$

The condition $1 \leq |x| \leq 2$ describes two symmetric closed intervals — one negative, one positive.

Final answer: $-2 \leq x \leq -1$ or $1 \leq x \leq 2$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 47 (ise4-047)

Difficulty: Challenge / Marks: 6

A particle is launched vertically and its height (in metres) above the ground after t seconds is given by $h(t) = 20t - 5t^2$. Find the values of t (with $t \geq 0$) for which the particle is more than 15 metres above the ground.

Worked solution.

Step 1: Set up the inequality from the model.

$$20t - 5t^2 > 15$$

"More than 15 metres above the ground" means the height function strictly exceeds 15.

Step 2: Rearrange to standard form.

$$-5t^2 + 20t - 15 > 0 \iff t^2 - 4t + 3 < 0$$

Divide both sides by -5 and flip the inequality. Dividing by a negative quantity always reverses the direction — a very common mistake to watch for.

Step 3: Factorise and solve.

$$t^2 - 4t + 3 = (t - 1)(t - 3) < 0 \implies 1 < t < 3$$

U-shape, strictly negative between the roots.

Step 4: Check the physical domain.

$$1 < t < 3 \text{ and } t \geq 0$$

The interval $(1, 3)$ already lies inside $t \geq 0$, so no additional restriction is needed. Times before launch are not physically meaningful and are excluded by the model anyway.

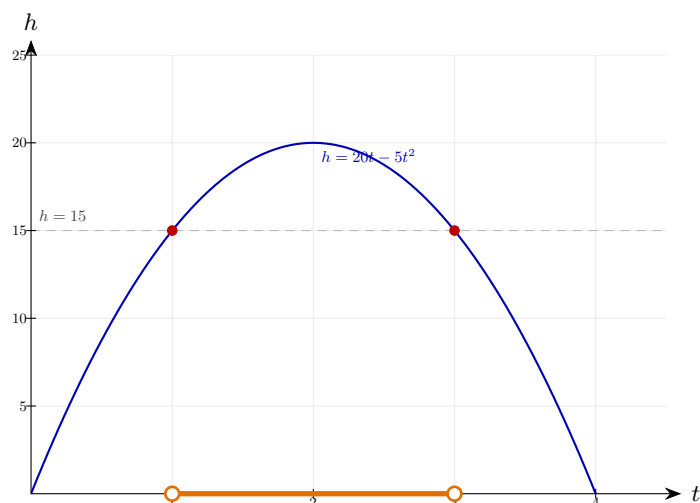
Step 5: State the answer with units.

$$1 < t < 3 \text{ (seconds)}$$

The particle is more than 15 metres above the ground between 1 and 3 seconds after launch.

Final answer: $1 < t < 3$ seconds

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 48 (ise4-048)

Difficulty: Challenge / Marks: 6

A company's daily profit (in pounds) from selling n units of a product is modelled by $P(n) = -n^2 + 80n - 700$. Find the values of n (with $n \geq 0$) for which the daily profit is at least £900.

Worked solution.

Step 1: Translate "at least £900" into an inequality.

$$-n^2 + 80n - 700 \geq 900$$

"At least" allows equality, so we use a non-strict $\geq$.

Step 2: Rearrange to standard form.

$$-n^2 + 80n - 1600 \geq 0 \iff n^2 - 80n + 1600 \leq 0$$

Subtract 900 from both sides, then multiply by -1 and reverse the inequality.

Step 3: Factorise.

$$n^2 - 80n + 1600 = (n - 40)^2$$

Discriminant $= 80^2 - 4(1)(1600) = 6400 - 6400 = 0$ — a repeated root at $n = 40$.

Step 4: Solve $(n - 40)^2 \leq 0$.

$$(n - 40)^2 \geq 0 \text{ always, with equality only at } n = 40$$

A real square cannot be negative, so $(n - 40)^2 \leq 0$ forces $(n - 40)^2 = 0$, giving $n = 40$.

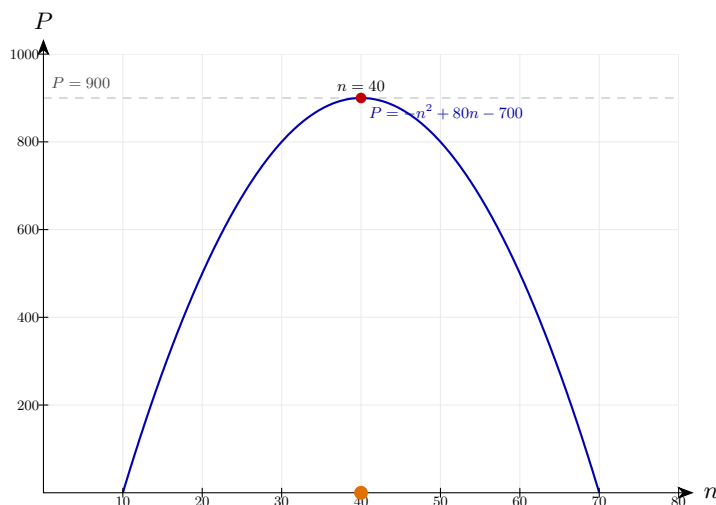
Step 5: Verify the physical domain.

$$n = 40 \geq 0 \checkmark$$

Selling 40 units yields exactly £900 profit, the maximum possible from this model. No other sales level achieves £900 or more.

Final answer: $n = 40$ units

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 49 (ise4-049)

Difficulty: Challenge / Marks: 6

The line $y = x + 1$ and the curve $y = x^2 - 5$ are sketched on the same axes. Find the set of values of x for which the line lies strictly above the curve.

Worked solution.

Step 1: Translate the geometric condition into an inequality.

$$x + 1 > x^2 - 5$$

The line is strictly above the curve where its y -value is strictly greater than the curve's y -value.

Step 2: Rearrange to standard form.

$$0 > x^2 - x - 6 \iff x^2 - x - 6 < 0$$

Bring all terms to one side; the x^2 coefficient is positive (we want it that way).

Step 3: Factorise.

$$x^2 - x - 6 = (x - 3)(x + 2)$$

Roots $x = -2$ and $x = 3$, which are the x -coordinates of the intersection points of the line and the curve.

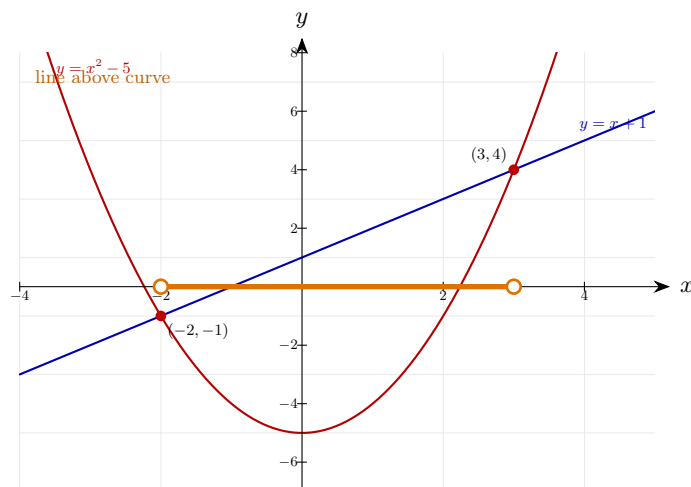
Step 4: Apply the U-shape rule.

$$(x - 3)(x + 2) < 0 \implies -2 < x < 3$$

Between the two intersection points the line sits above the curve; outside them the curve overtakes the line.

Final answer: $-2 < x < 3$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.

Quadratic Inequalities 50 (ise4-050)

Difficulty: Challenge | Marks: 7

The curve C has equation $y = x^2 - 6x + 11$ and the line ℓ has equation $y = 2x - 4$. (a) Show that the curve lies entirely above the x -axis. (b) Find the set of values of x for which the curve lies on or below the line.

Worked solution.

Step 1: (a) Show $x^2 - 6x + 11 > 0$ for all real x using the discriminant.

$$\Delta = (-6)^2 - 4(1)(11) = 36 - 44 = -8 < 0$$

The discriminant is negative, so the quadratic has no real roots and the parabola never crosses the x -axis.

Step 2: Combine with the sign of a .

$$a = 1 > 0, \Delta < 0 \implies x^2 - 6x + 11 > 0 \text{ for all } x$$

When $\Delta < 0$ the parabola is entirely on one side of the x -axis, taking the sign of a . Since $a > 0$, it is strictly above. (Alternatively, completing the square gives $(x - 3)^2 + 2 \geq 2 > 0$ for all x .)

Step 3: (b) Translate "curve on or below line" into an inequality.

$$x^2 - 6x + 11 \leq 2x - 4$$

The curve's y -value is at most the line's y -value.

Step 4: Rearrange to standard form.

$$x^2 - 8x + 15 \leq 0$$

Subtract $2x - 4$ from both sides and tidy.

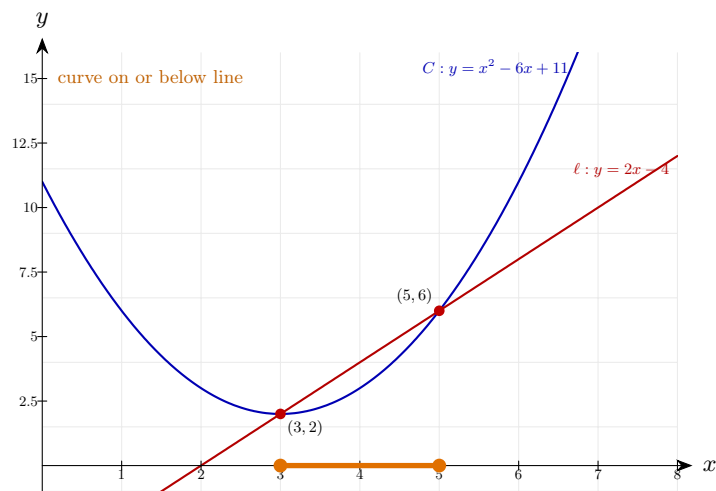
Step 5: Factorise and solve.

$$(x - 3)(x - 5) \leq 0 \implies 3 \leq x \leq 5$$

U-shape, non-positive on the closed interval between the roots. The endpoints $x = 3$ and $x = 5$ are the x -coordinates of the intersection points of C and ℓ , and are included because of $\leq$.

Final answer: (a) Shown: $\Delta < 0$ and $a > 0$, so the quadratic is positive for all x . (b) $3 \leq x \leq 5$

Diagram.



The parabola sketched, with the roots marked and the solution set highlighted on the x -axis.
